

Table S8. Model summaries for models filtered to only include threatened species that are within the prey range of cats, thus excluding 2 observations from 2 articles on *Eupleres goudotii* and *Megadyptes antipodes*. Model estimates $\pm$ 95% CIs and test statistics are reported along with sample sizes and residual unexplained heterogeneity (I^2), decomposed by hierarchical model structure (article and observation). The effect size used is given in the heading of each model (SMD=standardized mean difference or Hedges' g; Zr=correlation coefficient; lnOR=log odds ratio).

Term	Estimate	Moderator	Test statistics	Sample size	Residual heterogeneity
Abundance with/without cats All species Overall estimate SMD					
overall	-0.41 $\pm$ [-1.12, 0.3]	Overall estimate	$t_8 = -1.34, p = 0.22$	$N_{articles} = 9$ $N_{species} = 12$ $N_{observations} = 14$	$I^2_{total} = 58.7$ $I^2_{article} = 58.7$ $I^2_{obs} = 0$
Abundance before-after eradication All species Overall estimate SMD					
overall	-0.91 $\pm$ [-6.27, 4.45]	Overall estimate	$t_1 = -2.16, p = 0.28$	$N_{articles} = 2$ $N_{species} = 2$ $N_{observations} = 3$	$I^2_{total} = 26.5$ $I^2_{article} = 26.5$ $I^2_{obs} = 0$
Abundance on islands with/without cats All species Overall estimate lnOR					
overall	0.03 $\pm$ [-1.7, 1.76]	Overall estimate	$t_5 = 0.04, p = 0.97$	$N_{articles} = 6$ $N_{species} = 10$ $N_{observations} = 10$	$I^2_{total} = 42.1$ $I^2_{article} = 42.1$ $I^2_{obs} = 0$
Abundance spatial association All species Overall estimate Zr					
overall	0.1 $\pm$ [-1.25, 1.45]	Overall estimate	$t_3 = 0.24, p = 0.83$	$N_{articles} = 4$ $N_{species} = 2$ $N_{observations} = 4$	$I^2_{total} = 67.3$ $I^2_{article} = 33.6$ $I^2_{obs} = 34$
Abundance temporal association All species Overall estimate Zr					
overall	0.1 $\pm$ [-0.37, 0.57]	Overall estimate	$t_{10} = 0.47, p = 0.65$	$N_{articles} = 11$ $N_{species} = 12$ $N_{observations} = 18$	$I^2_{total} = 85$ $I^2_{article} = 56.6$ $I^2_{obs} = 28$
Reproduction with/without cats All species Overall estimate lnOR					
overall	-0.08 $\pm$ [-2.62, 2.47]	Overall estimate	$t_2 = -0.13, p = 0.91$	$N_{articles} = 3$ $N_{species} = 3$ $N_{observations} = 5$	$I^2_{total} = 68.8$ $I^2_{article} = 68.8$ $I^2_{obs} = 0$
Reproduction before-after eradication All species Overall estimate SMD					
overall	0.16 $\pm$ [-4.5, 4.82]	Overall estimate	$t_1 = 0.43, p = 0.74$	$N_{articles} = 2$ $N_{species} = 3$ $N_{observations} = 3$	$I^2_{total} = 75$ $I^2_{article} = 75$ $I^2_{obs} = 0$
Reproduction temporal association All species Overall estimate Zr					

Table S8. Model summaries for models filtered to only include threatened species that are within the prey range of cats, thus excluding 2 observations from 2 articles on *Eupleres goudotii* and *Megadyptes antipodes*. Model estimates $\pm$ 95% CIs and test statistics are reported along with sample sizes and residual unexplained heterogeneity (I^2), decomposed by hierarchical model structure (article and observation). The effect size used is given in the heading of each model (SMD=standardized mean difference or Hedges' g; Zr=correlation coefficient; lnOR=log odds ratio).

Term	Estimate	Moderator	Test statistics	Sample size	Residual heterogeneity
overall	0.09 $\pm$ [-0.62, 0.8]	Overall estimate	$t_3 = 0.41, p = 0.71$	$N_{articles} = 1$ $N_{species} = 1$ $N_{observations} = 4$	$I^2_{total} = 0$ $I^2_{article} = NA$ $I^2_{obs} = NA$
Abundance with/without cats Mammals Overall estimate SMD					
overall	-0.59 $\pm$ [-2.19, 1]	Overall estimate	$t_3 = -1.18, p = 0.32$	$N_{articles} = 4$ $N_{species} = 6$ $N_{observations} = 6$	$I^2_{total} = 60.4$ $I^2_{article} = 60.4$ $I^2_{obs} = 0$
Abundance on islands with/without cats Mammals Overall estimate lnOR					
overall	-0.25 $\pm$ [-20.55, 20.04]	Overall estimate	$t_1 = -0.16, p = 0.9$	$N_{articles} = 2$ $N_{species} = 4$ $N_{observations} = 4$	$I^2_{total} = 60.1$ $I^2_{article} = 60.1$ $I^2_{obs} = 0$
Abundance spatial association Mammals Overall estimate Zr					
overall	0.1 $\pm$ [-1.25, 1.45]	Overall estimate	$t_3 = 0.24, p = 0.83$	$N_{articles} = 4$ $N_{species} = 2$ $N_{observations} = 4$	$I^2_{total} = 67.3$ $I^2_{article} = 33.6$ $I^2_{obs} = 34$
Abundance temporal association Mammals Overall estimate Zr					
overall	0.19 $\pm$ [-0.32, 0.7]	Overall estimate	$t_8 = 0.87, p = 0.41$	$N_{articles} = 9$ $N_{species} = 10$ $N_{observations} = 16$	$I^2_{total} = 85.6$ $I^2_{article} = 56.2$ $I^2_{obs} = 29$
Abundance with/without cats Birds Overall estimate SMD					
overall	-0.29 $\pm$ [-3.46, 2.87]	Overall estimate	$t_2 = -0.4, p = 0.73$	$N_{articles} = 3$ $N_{species} = 4$ $N_{observations} = 6$	$I^2_{total} = 70.6$ $I^2_{article} = 70.6$ $I^2_{obs} = 0$
Abundance on islands with/without cats Birds Overall estimate lnOR					
overall	0.59 $\pm$ [-26.43, 27.6]	Overall estimate	$t_1 = 0.28, p = 0.83$	$N_{articles} = 2$ $N_{species} = 4$ $N_{observations} = 4$	$I^2_{total} = 75.6$ $I^2_{article} = 75.6$ $I^2_{obs} = 0$
Reproduction with/without cats Birds Overall estimate lnOR					

Table S8. Model summaries for models filtered to only include threatened species that are within the prey range of cats, thus excluding 2 observations from 2 articles on *Eupleres goudotii* and *Megadyptes antipodes*. Model estimates $\pm$ 95% CIs and test statistics are reported along with sample sizes and residual unexplained heterogeneity (I^2), decomposed by hierarchical model structure (article and observation). The effect size used is given in the heading of each model (SMD=standardized mean difference or Hedges' g; Zr=correlation coefficient; lnOR=log odds ratio).

Term	Estimate	Moderator	Test statistics	Sample size	Residual heterogeneity
overall	-0.08 $\pm$ [-2.62, 2.47]	Overall estimate	$t_2 = -0.13, p = 0.91$	$N_{articles} = 3$ $N_{species} = 3$ $N_{observations} = 5$	$I^2_{total} = 68.8$ $I^2_{article} = 68.8$ $I^2_{obs} = 0$
Reproduction before-after eradication Birds Overall estimate SMD					
overall	0.16 $\pm$ [-4.5, 4.82]	Overall estimate	$t_1 = 0.43, p = 0.74$	$N_{articles} = 2$ $N_{species} = 3$ $N_{observations} = 3$	$I^2_{total} = 75$ $I^2_{article} = 75$ $I^2_{obs} = 0$
Reproduction temporal association Birds Overall estimate Zr					
overall	0.09 $\pm$ [-0.62, 0.8]	Overall estimate	$t_3 = 0.41, p = 0.71$	$N_{articles} = 1$ $N_{species} = 1$ $N_{observations} = 4$	$I^2_{total} = 0$ $I^2_{article} = NA$ $I^2_{obs} = NA$